

A
Project Report
on
“ Online Library Management System ”



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Title of the Project

Online Library Management System (OLMS)

Problem Statement

Most of the universities ,colleges as well as school libraries in India still keep track of books, students and issue/return records using registers. The working of this kind of activities manually creates a lot of problems which takes a lot of time to find whether a particular book is available or not. It creates a lot of mistakes while writing entries, and records. Librarians also find it hard to know which books are overdue, how many copies of a book are left, or which student has not returned a book, because everything has to be checked by hand.

The Online Library Management System is being developed to solve these problems. It will act as a single digital platform where the librarian can add, update, or remove book records, and where students can search for books, check availability, and see their own borrowing history, all through a simple website. The main goal of this project is to reduce manual paperwork, save time, avoid human errors, and make the whole process of issuing and returning books fast, accurate, and easy to manage for both the librarian and the students. Hence, all the details will be available on a single click.

Abstract

The Online Library Management System is a web-based application designed to automate and simplify the complete management of library operations. It provides an efficient digital platform for maintaining records of books, authors, categories, registered users, issued books, returned books, and transaction history in a centralized database. Traditional library management often depends on manual registers and paper-based documentation, which increases the possibility of errors, duplicate entries, misplaced records, and delays in retrieving information. The proposed system overcomes these limitations by offering a secure, organized, and user-friendly environment where library information can be accessed quickly and accurately. Users can register, log in, search books using different criteria, check availability, and view complete book details without visiting multiple record sections. Librarians and administrators can efficiently add, update, delete, issue, and return books, manage user accounts, monitor due

dates, and maintain accurate transaction records. The centralized database improves consistency, reduces paperwork, saves time, minimizes human errors, and enhances the overall quality of library services for students, teachers, and administrators.

The project is developed using the Waterfall Software Development Life Cycle (SDLC) model, which follows a systematic sequence of requirement analysis, system design, implementation, testing, deployment, and maintenance. Each phase is completed before moving to the next, ensuring proper planning, documentation, and quality control throughout the development process. The system is designed to provide reliable performance, secure data management, and efficient handling of daily library activities. It enables faster access to information, better organization of library resources, and improved monitoring of issued and overdue books. By replacing manual procedures with an automated solution, the Online Library Management System increases operational efficiency, supports effective decision-making, and delivers a convenient experience for both users and librarians. The project also provides a strong foundation for future enhancements such as online book reservation, email notifications, barcode integration, and digital library services, making it a practical, scalable, and dependable solution for modern educational institutions.

Introduction

In a normal library management system, the activities involving maintenance of book records, borrowing and returning of books, members' data and calculating fines are done manually. Such activities are not only time consuming but also very ineffective in a situation where there are many books and users in the library. Searching for books, record keeping and report generation becomes difficult in such circumstances.

Another problem which is encountered in most libraries includes absence of records, improper inventory management, inability to monitor overdue books and also the lack of ability to offer services when the library is not working. Users have to physically visit the library in order to find out about availability and borrowing of the books.

The objective of creating the Online Library Management System is to solve such problems encountered in library management system in order to make the activities more convenient and easier. The system offers an automated platform

through which all the activities that are involved in the library can be carried out. The librarian will be able to maintain the book and member data, monitor transactions of the book, calculate fines and generate reports easily using the system. In addition, the user will be able to search for books, check their availability and reserve books among other activities through an online system.

The main objective of the system is to increase efficiency and effectiveness of the library management system.

Identification of Life Cycle Model with Justification

What We Want to Develop

We want to develop a Online based Library Management System. It will have four main modules:

- Book Management – Add, update, delete, and search books.
- Member Management – Register members and manage their details.
- Issue/Return Management – Issue books, return books, and calculate fines automatically for late returns.
- Reports – Show issued books, overdue books, and available books.

These are the basic features that almost every library must acquire. Since the requirements are clear and fixed before development starts, the **Waterfall Model** is the most suitable software development model for this project.

Life Cycle Model : Waterfall Model

For this project, the Waterfall Model has been selected as the Software Development Life Cycle (SDLC). The Waterfall Model is a linear, step-by-step approach in which one phase is completed fully before the next phase begins. The six phases followed in this project are: Feasibility Study , Requirement Analysis and Specification ,Design ,Coding and Unit Testing , Integration and System Testing , Maintenance, as shown in the diagram below :-

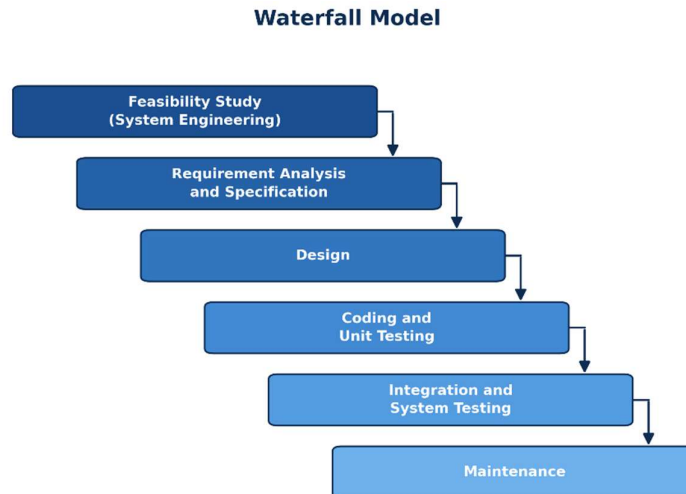


Fig : Waterfall Model showing the six sequential phases of OLMS development

Justification for Choosing the Waterfall Model

The waterfall model is a good choice for developing an Online Library Management System because the project requirements are simple and clear from the beginning. In this system, the main features such as student registration book management, book issue, book return, and fine calculation are already known. The waterfall model follows a step by step process, so each phase is completed before moving to the next phase. This makes the project easy to understand and manage especially for beginners. It also helps the development team plan the work properly. Testing is done after the coding is completed, which helps in finding and fixing errors before the system is used. Since the project does not require frequent changes this model works well. The documentation prepared in each phase is useful for future maintenance and updates. It also helps teachers, students, and developers understand the project easily. The waterfall model saves time by following a clear sequence of steps. It reduces confusion during development and improves project organization. Therefore the waterfall model is a suitable and reliable choice for developing an Online Library Management System.